has the same access. This approach is particularly valuable in environments where access depends not just on who someone is, but on what they're doing, when they're doing it, and under what conditions.

One of ABAC's core strengths is its ability to express complex policies in a centralized and maintainable way. Instead of creating numerous roles or custom access rules, organizations can develop reusable policies that reference attributes like clearance level, department, device status, or geolocation. As we've seen, policies can be layered and composed to reflect detailed requirements. Policies automatically adapt as input attributes change, allowing access decisions to be both accurate and resilient without manual updates.

For example, ABAC is well-suited for financial systems, where access to data must consider regulatory controls, time-based restrictions, and employee status. For instance, ACME Corp's finance team manages quarterly reports that are classified as confidential and region-specific, as shown in figure 6.4. Analysts with high clearance can access these reports during business hours, but outside those hours, access is limited to employees who are actively on call. Policies like these evaluate attributes from multiple sources—such as clearance level, department, region, and current shift status—to make access decisions in real time. Instead of manually defining roles for every possible scenario, ACME encodes these conditions as attribute-driven policies, providing precision and agility while easing the administrative burden.

However, that same flexibility brings new challenges. ABAC systems need to reliably retrieve and assess attributes from multiple sources, often in real-time. Policies—being code-like—can become difficult to write, manage, or debug without proper tools. Since decisions depend on a combination of inputs and logic, it can be hard to understand why access was granted or denied without good logging and tools to analyze logs. Table 6.5 highlights some of the most common strengths and limitations of ABAC in practice.